USB4 2.0 ENGINEERING CHANGE NOTICE FORM

Title: Disable Gen 4 Recovery on Asymmetric Transitions
Applied to: USB4 Specification Version 2.0

Brief description of the functional changes:
Since the default value of the Target Asymmetric Link might be different than Symmetric Link and Gen 4 Link Recovery flow checks this field to make sure it matched the actual Negotiated Link Width, we're removing the condition for a Disconnect in the Gen 4 Link Recovery flow when Target Asymmetric Link doesn't match the actual Link width and adding a Connection Manager note to Disable Gen 4 Link Recovery flow before doing Asymmetric Transitions.

Benefits as a result of the changes:
Removing a redundant requirement and moving the control to the Connection Manager.

An assessment of the impact to the existing revision and systems that currently conform to the USB specification:
None

An analysis of the hardware implications:
None

An analysis of the software implications:
None

An analysis of the compliance testing implications:
No need to check for Gen 4 Link Recovery flow when the Target Asymmetric Link doesn't match the Negotiated Link Width.

Actual Change

(a). Section 4.2.2.5
To Text:

CONNECTION MANAGER NOTE

The Connection Manager shall not initiate an Asymmetric transition on Inter-Domain Links.

The Connection Manager does the following before starting the transition from Symmetric Link to Asymmetric Link:

- *Disable CLx states on both sides of a Link. The Connection Manager may enable CLx states after the transition completes.*
- *Disable Gen 4 Link Recovery flow. The Connection Manager may enable Gen 4 Link Recovery flow after the transition completes.*
- *Set the Target Asymmetric Link field on both sides of the Link.*

The Connection Manager then sets the StartAsymmetricFlow field in the USB4 Port that switches from Tx1 to Rx2 to initiate the transition to an Asymmetric Link.

(b). Section 4.4.7 Gen 4 Link Recovery
To Text:

- If the *Enable Gen 4 Link Recovery* bit in PORT_CS_19 is set to 0b~~either of the following are true~~, initiate a Disconnect by driving SBTX to a logical low state for tDisconnectTx.~~:~~
  - ~~The value of the *Target Asymmetric Link* field does not reflect the actual negotiated Link as described in the *Negotiated Link Width* field.~~

    ~~The *Enable Gen 4 Link Recovery* bit in PORT_CS_19 is set to 0b~~.